2022-Summer Paper 3 – Subspecialities of Urology

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Q1. Describe various methods of Nephron sparing surgery in relation to Renal masses?

Type: LAQ (25 marks) (also answered in 2017-Summer Paper 1, Q1 and 2017-Winter Paper 4, Q3f)

Definition and rationale

- NSS (nephron-sparing surgery): complete removal of the tumour with maximal preservation of functioning parenchyma.
- RN (radical nephrectomy) causes CKD (chronic kidney disease), which raises cardiovascular events and death. PN (partial nephrectomy) gives equal cancer control with better function.
- Function after PN depends mainly on the **parenchyma preserved**, less on ischaemia time.

Indications

Category	Situations
Imperative (absolute)	Solitary kidney; bilateral tumours; RN would mean dialysis
Relative	Contralateral kidney threatened (stones, renal artery stenosis, diabetes, hypertension); hereditary RCC (renal cell carcinoma)
Elective	cT1a with a normal contralateral kidney: **standard of care**

- Now extended to cT1b–T2 when feasible (better function).
- **AUA (American Urological Association) 2021:** RN only if all three: high complexity; no CKD or proteinuria; normal contralateral kidney with eGFR (estimated glomerular filtration rate) likely >45 after RN.
- **Relative contraindications:** <20% of nephron mass left; pedicle encasement; venous tumour thrombus; cT4; cN1.

Preoperative planning

- Multiphase CT (computed tomography) or MRI (magnetic resonance imaging); chest imaging; eGFR, proteinuria.
- **R.E.N.A.L. score** (Radius, Exophytic, Nearness to collecting system, Anterior/posterior, Location to polar lines): 4–6 low, 7–9 moderate, 10–12 high. It predicts complications, ischaemia time and conversion. PADUA is an alternative.
- RMB (renal mass biopsy) when it changes management, and before (or at) ablation. 3D models shorten ischaemia (randomised trial).

Methods

Group	Methods
Surgical PN	Open, laparoscopic, robot-assisted (RAPN)
Excision technique	Standard (wedge) resection with a rim; enucleation; polar resection or heminephrectomy (≥30% of parenchyma)
Thermal ablation	Cryoablation, RFA (radiofrequency ablation), MWA (microwave ablation)
Developmental	SBRT (stereotactic body radiotherapy), HIFU (high-intensity focused ultrasound), irreversible electroporation

1. Open PN: steps

1. Flank extraperitoneal or transperitoneal approach.
2. **Early hilar control** of artery, vein and accessory vessels, even if off-clamp is planned.
3. Defat kidney; **intraoperative ultrasound** maps borders and depth.
4. Clamp; **ice slush** if ischaemia is expected ≥25–30 minutes (solitary kidney, CKD).
5. Excise with a thin rim; a negative margin is the priority, its width is not.
6. **Two-layer renorrhaphy:** inner layer closes vessels and collecting system; outer capsular sutures over a haemostatic bolster.
7. Unclamp, check haemostasis, close Gerota fascia, drain.

- Suits complex tumours in a solitary kidney.

2. Laparoscopic and robotic PN

- First laparoscopic PN 1993 (Winfield). **Transperitoneal** for anterior tumours (more space); **retroperitoneal** for posterior tumours (direct hilum, no bowel mobilisation).
- **Robotic vs laparoscopic:** shorter warm ischaemia, fewer conversions. **Robotic vs open:** less blood loss, shorter stay, fewer complications; same margins and eGFR.
- Haemostasis: sliding-clip renorrhaphy, barbed sutures, sealants.
- **Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes.

3. Enucleation vs standard resection

	Standard PN	Enucleation
Plane	Through normal parenchyma (thin rim)	Along tumour pseudocapsule
Pros	Fewer positive margins	Less bleeding, maximal parenchyma, eases zero ischaemia
Use	Default	Hereditary RCC (e.g. VHL [von Hippel–Lindau]), multifocal, solitary kidney, severe CKD

4. Vascular control strategies

Strategy	Verdict
En bloc or artery-only clamp	Equivalent; traditional warm ischaemia <25–30 minutes
Early unclamping (after inner layer)	May reduce pseudoaneurysm
Selective or super-selective segmental clamping	Avoids global ischaemia; no functional gain in a randomised trial
Off-clamp (zero ischaemia)	Small exophytic tumours; more blood loss; no long-term eGFR benefit in randomised trials
Parenchymal compression	Peripheral tumours
Hypothermia, mannitol	No benefit with a normal contralateral kidney

5. Thermal ablation

- Mostly **percutaneous** (CT or ultrasound guided). AUA 2021: alternative for **cT1a <3 cm**, especially elderly, comorbid, solitary kidney or VHL.
- **Cryoablation:** double freeze–thaw; cell death at ≤−40 °C; iceball 5–10 mm beyond the tumour. **RFA/MWA:** heat causes coagulative necrosis.
- Success = loss of enhancement on CT/MRI. Local recurrence is higher than PN but equal after repeat ablation.

Complications of PN

- Urine leak (drain, stent); haemorrhage, **pseudoaneurysm** or AV (arteriovenous) fistula (angioembolisation); acute kidney injury.
- Positive margin 0–7%: surveillance. Local recurrence 1–2% (elective), 3–5% (imperative).
- Hyperfiltration proteinuria when >50% of a solitary kidney is lost: ACE (angiotensin-converting enzyme) inhibitors.

Evidence

- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011):** RN vs PN, ≤5 cm: PN better function, similar oncology; 10-year overall survival 81% RN vs 76% PN; trial flawed.
- **Also know:** hereditary RCC treated at 3 cm; surveillance for frail patients.

Diagram to draw: clamped renal artery, tumour excised with rim, two-layer renorrhaphy.

> **Key points to remember:**
>
> - Imperative, relative, elective; elective PN is standard for cT1a.
> - Preserved parenchyma, not clamp time, governs function; trifecta warm ischaemia ≤25 min.
> - Enucleation for hereditary/multifocal; ablation for cT1a <3 cm after biopsy.
> - EORTC 30904: PN better function, similar cancer control.

Q2. Describe etiopathogenisis, clinical presentation, diagnosis and management of Renal TB?

Type: LAQ (25 marks) (also answered in 2018-Summer Paper 2, Q1)

Introduction

- Caused by the MTBC (*Mycobacterium tuberculosis* complex); *M. bovis* (and BCG [bacille Calmette–Guérin]) is innately resistant to pyrazinamide.
- UG (urogenital) TB (tuberculosis) is 15–20% of pulmonary TB in developing countries; second extrapulmonary site after lymph nodes. **Kidney is involved in 80%.**

Etiopathogenesis

- **Primary infection:** inhaled bacilli spread to hilar nodes, then by blood to the well-perfused kidney and epididymis, before immunity develops (up to 12 weeks).
- **Granuloma:** epithelioid cells and Langhans giant cells in a lymphocyte cuff, maintained by IFN-γ (interferon-γ). In 90–95% infection stays latent.
- **Reactivation** (lifetime 5–15%), even decades later: old age, diabetes, renal failure, HIV (human immunodeficiency virus), malnutrition, immunosuppression.
- **Routes:** haematogenous (principal); ascending (intravesical BCG, about 1%); contiguous (psoas, bowel).
- **Renal sequence:** bacilli lodge in **periglomerular capillaries** → cortical granulomas coalesce → **caseation** and cavities open into calyces → papillary necrosis, abscess, calcification → bacilluria seeds ureter and bladder.
- **Autonephrectomy:** caseo-cavernous type, or fibrotic calcified "putty" kidney.
- **Ureter and bladder (descending spread):** mural granulomas give strictures and a beaded or pipestem ureter. Cystitis starts around the ureteric orifices; after about a year of fibrosis the bladder contracts (<100 mL; thimble bladder <20 mL).
- **Key concept:** healing occurs by fibrosis, so infundibular and ureteric strictures (commonest at the **VUJ (vesicoureteric junction)**) and bladder contraction can worsen during treatment.
- **Also know:** granulomatous interstitial nephritis; miliary renal TB; amyloidosis; squamous metaplasia of the pelvis (squamous cell carcinoma risk); ESRD (end-stage renal disease) in about 7%.

Clinical presentation

- Renal TB is silent until it reaches the bladder; asymptomatic in about 8%.
- **Storage LUTS (lower urinary tract symptoms)** and dysuria (commonest), haematuria and flank pain (about one-third), colic <10% (sloughed papillae, clot, debris).
- Constitutional symptoms <20%. Recurrent "UTI (urinary tract infection)" not responding to antibiotics.
- Signs: renal angle tenderness, flank sinus, beaded vas, epididymal nodule; hypertension; renal failure if bilateral.

Diagnosis

- **Sterile pyuria** ± microscopic haematuria (>90%).
- **Urine AFB (acid-fast bacilli) culture: gold standard.** 3–5 first-void morning samples on consecutive days. LJ (Löwenstein–Jensen) solid medium 4–6 weeks; liquid MGIT (mycobacteria growth indicator tube) as little as 10 days. Gives drug susceptibility. Ziehl–Neelsen smear sensitivity <50%.

- **GeneXpert MTB/RIF:** result in 1–2 days; detects rifampicin resistance (MDR [multidrug-resistant] TB surrogate). Cannot monitor treatment.
- **LF-LAM (lateral-flow lipoarabinomannan):** point-of-care urine test in people with HIV (WHO [World Health Organization] 2019).
- **TST (tuberculin skin test)/IGRA (interferon-γ release assay):** show infection only, not active disease.
- **Chest radiograph in all** (pulmonary TB coexists in up to one-fifth).

Imaging	Findings
KUB (kidney–ureter–bladder) film	Calcification (24–44%); lobar "putty" kidney
IVU (intravenous urography) / CT (computed tomography) urography	**Moth-eaten calyx**, **phantom calyx** (infundibular stenosis), cavities, hydrocalycosis, **Kerr's kink**, pipestem or beaded corkscrew ureter, small bladder
Ultrasound	Follows hydronephrosis during treatment
MRU (magnetic resonance urography)	Children, pregnancy
Cystoscopy	**"Golf-hole" ureteric orifice**, ulcers; biopsy if culture negative

Management

Medical (all patients)

- **Multidrug therapy** because bacilli exist in different metabolic states and single drugs select resistance.
- Baseline liver and renal function, HIV, hepatitis, diabetes; DOT (directly observed therapy).
- **2HRZE/4HR:** isoniazid, rifampicin, pyrazinamide, ethambutol for 2 months, then isoniazid and rifampicin for 4 months; **pyridoxine** against isoniazid neuropathy. India's NTEP (National TB Elimination Programme) uses 2HRZE/4HRE.
- Complicated disease: some treat 12–24 months, or ≥6 months after surgery. The new 4-month rifapentine–moxifloxacin regimen is not used for UG TB.
- Toxicity: hepatitis (all except ethambutol); optic neuritis (ethambutol).
- **Monitor ultrasound** for worsening obstruction; steroids (anecdotal) for strictures.
- MDR-TB: second-line drugs (bedaquiline, pretomanid, linezolid).

Surgical (about half need it), after 4–6 weeks of ATT (antituberculous therapy)

Problem	Procedure
Obstruction	Early **DJ (double-J) stent** (succeeds in 41%) or **PCN (percutaneous nephrostomy)**
Localised polar lesion not responding	Partial nephrectomy
Non-functioning kidney + persistent disease or hypertension	**Nephrectomy** (laparoscopic preferred; ligate artery and vein separately)
PUJ (pelviureteric junction) stricture	Pyeloplasty; ureterocalicostomy
Lower ureteric stricture	Reimplantation, psoas hitch, Boari flap; ileal ureter if long
Contracted bladder	Augmentation; substitution if <20 mL

- Stricture not improving after 6 weeks of drugs → open repair.
- Contested: nephrectomy of a non-functioning kidney without active disease.

Follow-up

- Relapse 6.3–22% (mean about 5 years): urine culture and imaging for up to 10 years.

Diagram to draw: kidney showing caseous cavity, phantom calyx and VUJ stricture.

Key points to remember:

- Kidney 80%; periglomerular seeding; heals by fibrosis, so obstruction worsens on treatment.
- Sterile pyuria; 3–5 early-morning urine AFB cultures; GeneXpert for rifampicin resistance.
- 2HRZE/4HR with pyridoxine; surgery after 4–6 weeks of ATT.
- Relapse up to 22%; follow up 10 years.

Q3a. Assessment criteria for kidney donor.

Type: SAQ (10 marks) (also answered in 2017-Summer Paper 2, Q3b)

Assumed meaning: evaluation and acceptance criteria for a living kidney donor.

Principles

- **Donor safety first.** KDIGO (Kidney Disease: Improving Global Outcomes) 2017 is risk-based: lifetime ESKD (end-stage kidney disease) risk vs the centre's threshold.
- Multidisciplinary team with an **independent donor advocate**.
- India: **THOA (Transplantation of Human Organs Act) 1994, amended 2011:** near relatives; others need authorisation committee approval; swap donation allowed.

Assessment sequence

1. Compatibility: ABO group, HLA (human leukocyte antigen) typing, crossmatch, DSA (donor-specific antibody). Incompatible pair → paired exchange.

2. Kidney function (two methods, one measured, e.g. DTPA [diethylenetriamine pentaacetic acid] GFR)

Test	Accept	Individualise	Decline
GFR (glomerular filtration rate), mL/min/1.73 m²	≥90	60–89	<60
Albuminuria, ACR (albumin–creatinine ratio), mg/g	<30	30–100	>100

- Urinalysis; persistent microscopic haematuria → cystoscopy, imaging, ± biopsy.

3. Medical exclusions and limits

- **Diabetes:** type 1 excludes; HbA1c (glycated haemoglobin) and OGTT (oral glucose tolerance test) if at risk.
- **Hypertension:** controlled on 1–2 drugs without organ damage may be acceptable.
- **Obesity:** individualised risk.
- **Stones:** metabolic work-up; bilateral stones generally exclude.
- Active cancer, chronic infection, cardiovascular disease exclude.

4. Infection screen: HIV (human immunodeficiency virus), hepatitis B and C, CMV (cytomegalovirus), syphilis, tuberculosis.

5. Cardiopulmonary: ECG (electrocardiogram), chest X-ray, echo.

6. Age-appropriate cancer screening.

7. Anatomy: CT (computed tomography) angiography for vessel number, early branching, stones, masses. Split function if asymmetric: **the better kidney stays with the donor**; left preferred (longer vein).

8. Psychosocial: voluntary, no coercion or payment.

Counselling figures

- Function settles at 60–80% of baseline; mortality 3 per 10,000.
- **Muzaale (2014):** lifetime ESKD 90 per 10,000 donors vs 14 in matched healthy non-donors.

Also know: APOL1 (apolipoprotein L1) testing in African ancestry; more pre-eclampsia in later pregnancy.

Diagram to draw: donor work-up flowchart.

Key points to remember:

- KDIGO 2017: GFR ≥90 accept, <60 decline; ACR >100 decline.
- Type 1 diabetes and bilateral stones exclude.
- CT angiography; better kidney stays with the donor.
- THOA: near relatives, else authorisation committee.

Q3b. Management of Haemorrhagic cystitis.

Type: SAQ (10 marks)

Definition

- Diffuse inflammation and bleeding of the bladder mucosa, after excluding other sources (cystoscopy, cytology, upper tract imaging).
- **Exclude bladder cancer first:** cyclophosphamide and radiation are also risk factors for it.

Causes and mechanism

Cause	Mechanism
Cyclophosphamide, ifosfamide	Liver makes **acrolein**, excreted in urine, which sloughs urothelium; onset within 48 hours
Pelvic radiation (5%, 6 months to 10 years later)	**Obliterative endarteritis** → ischaemia, fibrosis, fragile neovessels
Viral	**BK polyomavirus**, adenovirus 11 and 35; after bone marrow or renal transplant
Others	Bacterial infection; busulfan, thiotepa; penicillins; amyloidosis

Prevention

- **Mesna** binds and inactivates acrolein; hyperhydration with forced diuresis, CBI (continuous bladder irrigation).

Management (escalate by severity; stability decides the branch)

```
Resuscitate, transfuse; correct coagulopathy; treat infection
→ large-bore catheter, CBI
→ cystoscopy under anaesthesia: clot evacuation, fulguration, biopsy
→ intravesical alum
→ bleeding after 48 h → hyperbaric oxygen (+ nephrostomies if bleeding continues)
→ formalin 1% → 4% (cystogram first)
→ unstable: internal iliac embolisation
→ cystectomy + conduit
```

Intravesical agent	Mechanism and method	Caution
Alum 1% (first line)	Protein precipitation, vasoconstriction; 200–300 mL/h; no anaesthesia	Aluminium toxicity in renal failure
Aminocaproic acid	Antifibrinolytic CBI, 200 mg/L	Only in a clot-free bladder; thromboembolism
Prostaglandins (carboprost), silver nitrate, hyaluronate	Vasoconstriction, coagulation, GAG (glycosaminoglycan) repair	Silver nitrate: cystogram
Formalin 1–4%	Protein fixation; **80–90% success**	Anaesthesia; **cystogram to exclude reflux**; fibrosis, small bladder, ureteric stricture

- **HBOT (hyperbaric oxygen therapy):** 100% oxygen at 2–3 atmospheres, 30–40 sessions; neovascularisation; 80–90% response, but many relapse. **RICH-ART** trial: better urinary symptom score in radiation cystitis.
- **Angioembolisation:** anterior division of internal iliac; spare the posterior division (gluteal pain).

- **Nephrostomies:** divert urokinase-rich urine.
- **Cystectomy with diversion:** last resort; remove the bladder (pyocystis risk).
- **Viral:** supportive; cidofovir (intravenous or intravesical).

Also know: tranexamic acid and laser ablation are unproven; oral pentosan polysulfate in children.

Diagram to draw: stepwise escalation ladder.

> **Key points to remember:**
> - Acrolein (prevent with mesna); radiation endarteritis; BK virus.
> - Alum first; formalin only after cystogram.
> - Hyperbaric oxygen for radiation cystitis (RICH-ART); embolise if unstable.

Q3c. Mini Perc.

Type: SAQ (10 marks)

See 2022-Summer Paper 1, Q1 (full answer).

Key points to remember (SAQ angle):

- Definition by sheath size: mini-PCNL (percutaneous nephrolithotomy) 15–22 Fr; ultra-mini 11–14 Fr; micro-perc <10 Fr (standard 24–34 Fr).
- Main advantage: less bleeding and fewer transfusions, shorter stay, with stone-free rates non-inferior to standard PCNL.
- Main limitation: longer operating time and higher intrarenal pressure (fever, sepsis); the advantage disappears against a 24 Fr tract.

Q3d. Methods of preoperative bowel preparation.

Type: SAQ (10 marks)

Rationale

- Bowel used in urinary diversion is opened into a sterile urinary tract. Faecal spillage causes wound infection, abscess, leak and sepsis.
- Flora: aerobes (*E. coli*, *Enterococcus faecalis*); anaerobes (*Bacteroides*, *Clostridium*).

Segment	Organisms per gram
Jejunum	$10–10^5$
Distal ileum	$10^5–10^7$
Ascending colon	$10^6–10^8$
Descending colon	$10^{10}–10^{12}$

Methods

1. Dietary: clear liquids the day before; clear drinks until 2 hours before surgery.

2. Mechanical bowel preparation (MBP) reduces faecal **amount**, so total bacteria, not concentration.

- **PEG (polyethylene glycol)–electrolyte** lavage: non-absorbable; preferred in elderly, renal or heart failure, cirrhosis.
- Sodium phosphate, magnesium citrate; enemas.
- Historical: 3-day purgation, whole-gut irrigation by nasogastric tube (9–12 L saline), mannitol (feeds bacteria).
- Harms: dehydration, electrolyte loss; liquid stool may spill more.

3. Oral antibiotic preparation reduces bacterial **concentration**.

- **Neomycin + erythromycin base** (Nichols–Condon, 1977: septic complications 43% with MBP alone vs 9% with added antibiotics); neomycin + metronidazole; kanamycin.
- Harms: diarrhoea, *C. difficile* colitis, candidiasis; useless in obstruction.

4. Intravenous prophylaxis (most important): anaerobic plus aerobic cover (e.g. cephalosporin + metronidazole) within 1–2 hours before incision; it must be in tissue before contamination.

5. ERAS (enhanced recovery after surgery): no routine MBP; prehabilitation, multimodal opioid-sparing analgesia, chewing gum, **alvimopan** (peripheral μ-opioid antagonist, 12 mg before surgery), early feeding and mobilisation.

Evidence

- Colorectal meta-analyses and Cochrane: MBP does not reduce leak or abscess; may raise SSI (surgical site infection).
- Oral antibiotics added to MBP reduce SSI and leaks in large colectomy series.
- Small cystectomy trials: no difference without preparation.

Who still needs full preparation

- Salvage cystectomy after radiation, or cystectomy for benign disease (radiation cystitis, fistula, neurogenic bladder); immunosuppression.

Also know: wound protectors reduce SSI; enema before transrectal biopsy.

Diagram to draw: bacterial load along the gut.

Key points to remember:

- Mechanical lowers amount; antibiotics lower concentration.
- Nichols–Condon: neomycin + erythromycin.
- Intravenous antibiotics before incision are essential; ERAS omits routine MBP.

Q3e. IPSS score

Type: SAQ (10 marks)

Definition

- **IPSS (International Prostate Symptom Score):** validated, self-administered questionnaire grading male LUTS (lower urinary tract symptoms) over the past month.
- Based on the **AUA-SI (American Urological Association Symptom Index, 1992)**; adopted by WHO (World Health Organization) in 1993 by adding one quality-of-life question.

Items

No.	Symptom	Phase
1	Incomplete emptying	Voiding
2	Frequency (again within 2 hours)	Storage
3	Intermittency	Voiding
4	Urgency	Storage
5	Weak stream	Voiding
6	Straining	Voiding
7	Nocturia	Storage

- **Scoring:** items 0–5. 0 not at all; 1 less than 1 time in 5; 2 less than half the time; 3 about half; 4 more than half; 5 almost always. **Nocturia scored by number of times** (0 to ≥5).
- **Total 0–35:** voiding subscore 4 items, storage 3 (not validated).

Severity	Score
Mild	0–7
Moderate	8–19
Severe	20–35

- **QoL (quality of life)/bother question:** "If you were to spend the rest of your life with your urinary condition as it is now, how would you feel?" Scored 0 (delighted) to 6 (terrible); reported separately, outside the 35.

Uses

- First step in LUTS work-up (AUA and EAU [European Association of Urology]); grades severity and bother.
- Guides treatment: mild, non-bothersome → watchful waiting; bothersome → medical therapy or surgery.
- Monitors response: **a 3-point fall is the minimum perceptible change**.
- Standard end point in trials.

Limitations

- **Not disease-specific:** women score similarly; does not diagnose BOO (bladder outlet obstruction) or BPH (benign prostatic hyperplasia); correlates poorly with urodynamics, prostate size and flow.
- No incontinence, pain or post-micturition items.
- Needs literacy.

Alternatives

- **VPSS (Visual Prostate Symptom Score):** pictures, for low literacy.
- BPH Impact Index; ICIQ-MLUTS (International Consultation on Incontinence Questionnaire–Male LUTS); DAN-PSS (Danish Prostate Symptom Score); OAB-q (overactive bladder questionnaire); LURN (Lower Urinary Tract Research Network) SI-29 adds incontinence and pain.

Diagram to draw: IPSS chart with scoring columns.

> **Key points to remember:**
> - 7 items × 0–5 = 0–35; bands 0–7, 8–19, 20–35.
> - QoL 0–6 is separate; IPSS = AUA-SI + QoL.
> - 3-point change is noticeable; IPSS does not diagnose BOO.

Q3f. Isolated penopubic Epispadias.

Type: SAQ (10 marks)

Definition

- Part of the **exstrophy–epispadias complex**: a defect of the dorsal urethral wall, replaced by a broad dorsal mucosal strip (urethral plate).
- Male classification by meatus: glanular, penile, **penopubic** (subsymphyseal). About **70% of male cases are complete (penopubic)**, with incontinence.
- Incidence of isolated male epispadias about 1 in 117,000.

Embryology

- Absent mesodermal reinforcement of the cloacal membrane; genital tubercles fuse **caudal** to it, giving a dorsal urethral defect.

Clinical features

- Penopubic meatus; outlet may admit a finger → **total incontinence**.
- **Short, broad, upturned penis with dorsal chordee**, like exstrophy.
- **Pubic diastasis**, less than in exstrophy.
- Small bladder (no outlet resistance); **VUR (vesicoureteric reflux) 30–40%** in males.
- **Female (Davis retrosymphyseal type):** bifid clitoris, incontinence.

Evaluation

- Renal ultrasound; VCUG (voiding cystourethrogram); cystoscopy and capacity under anaesthesia.
- **Key decision:** epispadias or exstrophy pathway. Treat as exstrophy (osteotomy, closure) if pubic diastasis >3 cm, poor intrasymphyseal bar, absent sphincter or tiny bladder.

Management

- **Aims:** continence, safe upper tracts, a straight functional penis.
- **Preoperative testosterone** (2 mg/kg intramuscular, 5 and 2 weeks before).

Stage 1: urethroplasty and penile reconstruction (6–12 months)

- Release chordee, divide suspensory ligaments, free corpora from inferior rami.
- **Modified Cantwell–Ransley:** plate tubularised and moved ventral to the corpora, which are rotated medially (± cavernocavernostomy).
- **Mitchell–Bägli complete penile disassembly:** reliable chordee correction, but often leaves hypospadias.
- Outlet resistance rises, so capacity grows.

Stage 2: bladder neck reconstruction (4–5 years, capacity about 100 mL)

- **Young–Dees–Leadbetter** bladder neck repair + Marshall–Marchetti–Krantz suspension + **ureteric reimplantation**. Osteotomy usually unnecessary.
- **Good capacity:** single-stage repair possible.

Outcomes and complications

- Continence after bladder neck reconstruction: **males 47–82%**, females 70–87%; capacity is the main predictor.
- Fistula (up to about one-fifth early), stricture rare; erections and paternity usually preserved.
- Better than exstrophy: bladder never exposed or scarred.

- Failure: injectables, augmentation.

Diagram to draw: dorsal penopubic meatus and plate.

> **Key points to remember:**
> - About 70% of male epispadias are penopubic and incontinent; VUR 30–40%.
> - Diastasis >3 cm → exstrophy pathway.
> - Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL.

Rapid Revision Sheet – 2022-Summer Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- Sterile pyuria; 3–5 early-morning urine AFB cultures; GeneXpert for rifampicin resistance.
- 2HRZE/4HR with pyridoxine; surgery after 4–6 weeks of ATT.
- Relapse up to 22%; follow up 10 years.

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- CT angiography; better kidney stays with the donor.
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- Acrolein (prevent with mesna); radiation endarteritis; BK virus.
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